

# Vaibhav Saini

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## Education

**G.B. Pant DSEU Campus-I**, Delhi, India

Aug 2024 - Jun 2028

Bachelor of Technology in Computer Science and Engineering

**Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

## Technical Skills

**Languages:** TypeScript, JavaScript, Python, SQL, Dart

**Frontend:** Next.js, React, Tailwind CSS, Flutter

**Backend:** Node.js, Express.js, NestJS, FastAPI, tRPC, Celery

**Databases & ORMs:** PostgreSQL, Prisma, Drizzle ORM, Supabase, MongoDB

**Auth & Security:** JWT, Role-Based Access Control (RBAC), bcrypt

**Tools:** Git, Docker, Postman, Cypress, Swagger, LangGraph

## Projects

**Buildra** - AI Code Generation Platform

GitHub

*Next.js, TypeScript, Inngest, E2B, Gemini AI, Prisma, PostgreSQL*

- Architected an AI agent loop using Gemini that iteratively generates, validates, and self-corrects Next.js application code across multiple files.
- Integrated E2B cloud sandboxes for secure execution of AI-generated code with real-time preview and file exploration.
- Built event-driven pipeline with Inngest for reliable multi-step AI orchestration with automatic retries and step-level observability.
- Implemented Server-Sent Events for token-by-token streaming, reducing perceived response latency by approximately 60%.

**Orbit** - Multi-Tenant Collaborative Platform

GitHub

*Next.js, tRPC, Drizzle ORM, PostgreSQL, Supabase Auth*

- Designed multi-tenant PostgreSQL schema with organization-scoped RBAC and denormalized orgId for query performance while maintaining strict data isolation.
- Implemented tRPC with middleware for end-to-end type safety between client and server, eliminating runtime type errors.
- Built fractional indexing system for Kanban board enabling O(1) drag-and-drop reordering without rewriting adjacent database rows.
- Created secure invitation system with org-scoped invite links and automatic role assignment on join.

**StackAudit** - AI-Powered SaaS Cost Optimizer

GitHub

*Next.js, TypeScript, Supabase, Claude API, OpenAI API*

- Built rules-based audit engine generating 12–15 specific cost optimization recommendations per report.
- Designed resilient LLM summary pipeline with three-tier degradation: Claude → OpenAI → template fallback.
- Implemented shareable public report URLs with auto-generated OpenGraph metadata for social preview.
- Created lead-capture funnel with email collection, demonstrating product and go-to-market thinking.

## Certifications

**Google Cloud Skills Boost — Vertex AI: Prompt Design**

Jul 2025